

Notes: Buera, Kaboski & Shin (AER, 2011)

Finance and Development: A Tale of Two Sectors

Contents

1	Big picture	3
2	Incremental contribution of this paper	3
2.1	What this paper adds relative to earlier finance-development work	3
2.2	New conceptual contribution: scale as the bridge from microfinance to macro development . .	4
2.3	New quantitative contribution: self-financing and dynamic wealth accumulation	4
2.4	New empirical contribution: sectoral scale facts and Mexico–U.S. validation	4
3	Empirical facts and measurement	4
3.1	Fact 1: relative prices and development	4
3.2	Fact 2: sectoral relative productivity	5
3.3	Fact 3: financial development and relative prices	5
3.4	Fact 4: scale differences across sectors	5
4	Model: primitives and decisions	5
4.1	Environment	5
4.2	Preferences	6
4.3	Entrepreneurial production	6
4.4	Unconstrained factor demands	6
5	Financial frictions and individual problem	7
5.1	Credit and rental markets	7
5.2	Enforceability condition	7
5.3	Rental limit	7
5.4	Occupation choice	7
5.5	Worker problem	7
5.6	Entrepreneur problem	8
6	Equilibrium, calibration, and quantitative design	8
6.1	Stationary competitive equilibrium	8
6.2	Perfect-credit benchmark	8
6.3	Calibration strategy	8
6.4	Financial development exercise	9
7	Main quantitative results	9
7.1	Aggregate income and TFP	9
7.2	Capital-to-output ratios and investment prices	9
7.3	Sector-level productivity	9
7.4	Decomposition of sectoral TFP losses	9
7.5	Establishment-size predictions	10
8	Tables and figures	10
8.1	Figure 1 and Table 1: motivating facts	10

8.2	Table 2 and Figure 2: calibration and U.S. establishment distribution	10
8.3	Figure 3: aggregate effects of financial frictions	11
8.4	Figure 4: sector-level TFP decomposition	11
8.5	Figure 5: relative prices and relative productivity versus finance	12
8.6	Figure 6 and Figure 7: establishment-level evidence	12
9	Short summary	13
10	Conclusion	13
10.1	Core conclusion	13
10.2	Economic intuition	13
10.3	Incremental contribution takeaway	14
10.4	Takeaway for readers	14

1 Big picture

- **Question.** How much can financial frictions explain cross-country differences in income, aggregate productivity, sectoral productivity, relative prices, and investment rates?
- **Core mechanism.** Financial frictions prevent high-talent but poor entrepreneurs from renting enough capital and entering the appropriate sector. They also keep low-talent but wealthy entrepreneurs in operation. This misallocates capital and entrepreneurial talent.
- **Two-sector idea.** The economy has a small-scale sector, interpreted as services, and a large-scale sector, interpreted as manufacturing. Manufacturing has higher fixed operating costs and therefore larger average establishments.
- **Why sectoral scale matters.** Larger-scale technologies require larger financing. Therefore, financial frictions bind more severely in manufacturing than in services.
- **Main quantitative result.** Financial frictions can explain a large share of cross-country income and TFP differences. They can generate about a factor-of-two difference in output per worker across financial-development levels.
- **Sectoral result.** Manufacturing TFP falls more than service TFP in financially underdeveloped economies. This generates higher relative prices of manufactured goods to services.
- **Capital-output result.** Poor financial systems can generate low capital-output ratios at common prices, mainly because financial frictions make investment goods relatively expensive.
- **Decomposition result.** In services, the TFP loss is mostly capital misallocation. In manufacturing, the TFP loss is mostly talent misallocation and distorted entry/exit.
- **Empirical validation.** The model predicts too few and too large manufacturing establishments but too many and too small service establishments in financially underdeveloped economies. A Mexico–United States case study supports this prediction.
- **Bottom line.** Financial development affects aggregate productivity not only by changing capital accumulation but by changing which entrepreneurs operate, how much capital they use, and which sectors expand.

2 Incremental contribution of this paper

2.1 What this paper adds relative to earlier finance-development work

- Earlier empirical work documents that financial development is strongly correlated with income per capita, growth, and productivity. This paper builds a quantitative general-equilibrium mechanism that maps financial frictions into those macro facts.
- Earlier micro work documents credit constraints and capital misallocation. This paper embeds those frictions in a two-sector development model with endogenous occupational choice, savings, and entry.
- Earlier quantitative models of financial frictions often use one-sector economies. This paper shows that sectoral scale differences are essential because financing needs differ across sectors.
- Earlier models often emphasize capital misallocation only. This paper decomposes the impact into capital misallocation, talent misallocation, and distortions in the number of establishments.

Interpretation: The paper’s contribution is not simply saying “finance matters.” It explains *how* finance matters differently across sectors and why this difference shows up in relative prices, relative productivity, and establishment-size distributions.

2.2 New conceptual contribution: scale as the bridge from microfinance to macro development

- The paper identifies establishment scale as a key state variable connecting micro production technology to macro development.
- Sectors with larger per-period fixed costs require larger operating scales.
- Large-scale sectors are more vulnerable to collateral constraints because productive entrepreneurs need more external finance before they can operate efficiently.
- This gives a mechanism for why poor countries are especially unproductive in manufacturing and investment-goods sectors.

Economic message: Sectoral scale differences transform financial frictions from a generic distortion into a source of systematic relative productivity and relative price patterns across sectors.

2.3 New quantitative contribution: self-financing and dynamic wealth accumulation

- Individuals can save over time and partially overcome borrowing limits.
- This self-financing channel is essential because it prevents the model from overstating the impact of financial frictions.
- A two-period overlapping-generations comparison shows that shutting down long-horizon self-financing exaggerates the aggregate effect of financial frictions by as much as 50 percent.
- The model therefore treats financial frictions as dynamic barriers rather than static borrowing caps.

Interpretation: The model's forward-looking savings channel is a key improvement over simpler static credit-constraint models. It explains why financial frictions are severe but not infinitely severe: talented entrepreneurs can eventually accumulate internal funds, although this is harder in large-scale sectors.

2.4 New empirical contribution: sectoral scale facts and Mexico–U.S. validation

- The paper documents that manufacturing establishments are much larger than service establishments in the United States and OECD data.
- It links sectoral scale to external finance dependence and capital shares.
- It then tests a model prediction using disaggregated Mexico–United States establishment-size data.
- The evidence supports the model's prediction that Mexico has relatively larger establishments in large-scale manufacturing industries and relatively smaller establishments in service industries.

Economic message: The paper moves beyond aggregate correlations by testing a sharp sectoral prediction: financial underdevelopment changes the relative size distribution of establishments across sectors.

3 Empirical facts and measurement

3.1 Fact 1: relative prices and development

- The paper uses Penn World Table and International Comparison Program data to study the relative price of manufactured goods to services.
- Poor countries have high relative prices of manufactured goods to services.

- This pattern is interpreted as evidence that poorer countries are especially inefficient at producing manufactured goods.

Interpretation: If relative prices reflect sectoral productivity differences, high manufacturing-service relative prices imply relatively low manufacturing productivity in poor countries.

3.2 Fact 2: sectoral relative productivity

- The paper uses sector-level productivity data from OECD countries to compare manufacturing TFP to service TFP.
- The relative TFP of manufacturing to services is positively related to GDP per worker.
- Although the sample is smaller than the relative-price sample, the pattern supports the relative-price evidence.

Interpretation: The sectoral productivity data suggest that manufacturing is especially weak in less developed economies, consistent with the model's large-scale-sector mechanism.

3.3 Fact 3: financial development and relative prices

- Financial development is measured by the external-finance-to-GDP ratio.
- Countries with lower external finance have higher relative prices of manufactured goods to services.
- Countries with higher external finance tend to have higher manufacturing-service relative productivity.

Interpretation: The empirical relationship between finance and relative prices is the model's main macro-sectoral target.

3.4 Fact 4: scale differences across sectors

- Table 1 shows that manufacturing establishments are much larger than service establishments.
- In U.S. data, manufacturing has 47 workers per establishment and services have 14.
- In OECD enterprise data, manufacturing has 28 workers per enterprise and services have 8.
- Manufacturing also has higher external dependence and a larger capital share.

Interpretation: Manufacturing has larger scale and larger financing needs. This is the empirical basis for treating manufacturing as the large-scale sector.

4 Model: primitives and decisions

4.1 Environment

- There are two sectors: S for small-scale services and M for large-scale manufacturing.
- Services are used only for consumption.
- Manufactured goods are used for consumption and investment and are the numeraire.
- Individuals are infinitely lived and heterogeneous in wealth a and entrepreneurial talent $z = (z_S, z_M)$.
- Entrepreneurial ideas evolve stochastically. A current idea survives with probability γ and is redrawn with probability $1 - \gamma$.

Interpretation: The stochastic talent process creates a continual need to reallocate capital and labor toward currently productive entrepreneurs. Financial frictions slow down this reallocation.

4.2 Preferences

Individual preferences are

$$U(c) = E \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (1)$$

where

$$u(c_t) = \frac{1}{1-\sigma} \left[\psi c_{S,t}^{1-1/\varepsilon} + (1-\psi) c_{M,t}^{1-1/\varepsilon} \right]^{\frac{1-\sigma}{1-1/\varepsilon}}. \quad (2)$$

- β is the discount factor.
- σ is relative risk aversion.
- ε is the intratemporal elasticity of substitution between services and manufactured goods.
- ψ governs the service consumption share.

Interpretation: Non-homotheticities are not the focus. The central role of preferences is to determine demand for services versus manufactured goods and the equilibrium relative price.

4.3 Entrepreneurial production

An entrepreneur in sector $j \in \{S, M\}$ with talent z_j operates one establishment and pays a per-period fixed cost κ_j . Production after paying the fixed cost is

$$y_j = z_j k^\alpha l^\theta, \quad \alpha + \theta < 1. \quad (3)$$

The fixed cost satisfies

$$\kappa_M > \kappa_S. \quad (4)$$

The static profit before borrowing constraints is

$$\pi_j(k, l; R, w, p) = p_j z_j k^\alpha l^\theta - Rk - wl - (1+r)p_j \kappa_j, \quad (5)$$

with $p_M = 1$.

Interpretation: Fixed costs generate nonconvexity and scale. Manufacturing has a higher fixed cost, so efficient manufacturing establishments are larger and require more external finance.

4.4 Unconstrained factor demands

The unconstrained static demands are

$$(k_j^u(z_j), l_j^u(z_j)) = \arg \max_{k,l} \{p_j z_j k^\alpha l^\theta - Rk - wl\}. \quad (6)$$

Interpretation: The gap between constrained and unconstrained capital demand is the main measure of capital misallocation in the model.

5 Financial frictions and individual problem

5.1 Credit and rental markets

- Financial intermediaries accept deposits, lend the fixed cost, and rent capital to entrepreneurs.
- Intermediaries make zero profit, so the rental rate satisfies

$$R = r + \delta. \quad (7)$$

- Financial contracts are imperfectly enforceable.
- After production, an entrepreneur can default and keep a fraction $1 - \phi$ of undepreciated capital and output net of labor costs.
- The parameter $\phi \in [0, 1]$ indexes contract enforcement: $\phi = 0$ is no enforceability; $\phi = 1$ is perfect enforceability.

Interpretation: Financial development is summarized by ϕ . Better financial institutions make higher capital rentals enforceable.

5.2 Enforceability condition

Capital rental k in sector j is enforceable if and only if

$$\max_l \{p_j z_j f(k, l) - wl\} - Rk - (1 + r)p_j \kappa_j + (1 + r)a \geq (1 - \phi) \left[\max_l \{p_j z_j f(k, l) - wl\} + (1 - \delta)k \right]. \quad (8)$$

Interpretation: The left-hand side is the entrepreneur's payoff from honoring contracts. The right-hand side is the payoff from default. More wealth a , higher talent z_j , and stronger enforcement ϕ relax the constraint.

5.3 Rental limit

The enforceability condition implies a rental limit

$$k \leq \bar{k}_j(a, z_j; \phi), \quad (9)$$

where $\bar{k}_j$ is increasing in wealth, talent, and enforcement.

Interpretation: The key financial friction is not a fixed borrowing cap. It is an endogenous collateral constraint that depends on wealth, talent, and sectoral technology.

5.4 Occupation choice

Each period, individuals choose whether to work or operate an establishment in either sector:

$$v(a, z) = \max\{v_W(a, z), v_S(a, z), v_M(a, z)\}. \quad (10)$$

Interpretation: Occupational choice is central to talent misallocation. A talented poor individual may be forced to work or enter the wrong sector because he lacks collateral.

5.5 Worker problem

A worker chooses consumption $c = (c_S, c_M)$ and next-period wealth a' :

$$v_W(a, z) = \max_{c, a' \geq 0} \{u(c) + \beta [\gamma v(a', z) + (1 - \gamma) E_{z'} v(a', z')]\} \quad (11)$$

subject to

$$p \cdot c + a' \leq w + (1 + r)a. \quad (12)$$

Interpretation: Workers may save in order to finance future entrepreneurial opportunities.

5.6 Entrepreneur problem

An entrepreneur in sector j solves

$$v_j(a, z) = \max_{c, a', k, l \geq 0} \{u(c) + \beta [\gamma v(a', z) + (1 - \gamma) E_{z'} v(a', z')]\} \quad (13)$$

subject to

$$p \cdot c + a' \leq p_j z_j f(k, l) - Rk - wl - (1 + r)p_j \kappa_j + (1 + r)a, \quad (14)$$

and

$$k \leq \bar{k}_j(a, z_j; \phi). \quad (15)$$

Interpretation: Entrepreneurs choose both production scale and savings. The savings decision is crucial because self-financing can gradually relax future constraints.

6 Equilibrium, calibration, and quantitative design

6.1 Stationary competitive equilibrium

A stationary equilibrium consists of prices, policy functions, an invariant wealth-talent distribution, and rental limits such that:

- individuals optimize;
- intermediaries make zero profits;
- rental limits satisfy enforceability;
- capital, labor, service goods, and manufacturing goods markets clear;
- the distribution of wealth and talents is stationary under optimal policies and stochastic talent transitions.

Interpretation: The equilibrium jointly determines income distribution, occupational selection, sectoral output, capital accumulation, and relative prices.

6.2 Perfect-credit benchmark

The perfect-credit benchmark sets

$$\bar{k}_j(a, z_j; \phi) = k_j^u(z_j) \quad (16)$$

for all entrepreneurs.

Interpretation: The perfect-credit economy isolates the technological allocation without borrowing constraints. It is calibrated to match U.S. data and serves as the benchmark for financial development.

6.3 Calibration strategy

- The perfect-credit model is calibrated to U.S. establishment data and macro moments.
- Target moments include employment concentration, earnings concentration, average establishment size in services and manufacturing, establishment exit rates, manufacturing's share of GDP, and the real interest rate.
- Sector-specific fixed costs are chosen to match average sectoral establishment sizes.
- The tail parameter of entrepreneurial talent is chosen to match the thick right tail in the establishment-size distribution.

Interpretation: The calibration is not targeted directly to cross-country productivity gaps. It first matches U.S. micro establishment facts, then changes financial development to study cross-country implications.

6.4 Financial development exercise

The paper varies ϕ from no enforceability to perfect credit and maps it into the endogenous external-finance-to-GDP ratio. All cross-country predictions are then plotted against external finance to GDP.

Interpretation: The financial development measure in the model is endogenous. This allows model outcomes to be compared to country data using the same external-finance-to-GDP axis.

7 Main quantitative results

7.1 Aggregate income and TFP

- Financial frictions reduce output per worker by almost a factor of two between the lowest and highest financial-development economies in the model.
- The model explains about 80 percent of the U.S.–Mexico output-per-worker gap.
- Aggregate TFP falls by almost 40 percent in the least financially developed economy relative to the perfect-credit benchmark.
- Most of the output loss is due to lower TFP rather than lower factor accumulation.

Economic message: Financial frictions are productivity distortions, not merely investment distortions.

7.2 Capital-to-output ratios and investment prices

- Financial frictions reduce the capital-output ratio measured at common prices.
- Investment rates measured at domestic equilibrium prices are roughly stable across financial development.
- The low common-price capital-output ratio is mainly driven by high relative prices of manufactured investment goods.

Interpretation: Financial underdevelopment makes investment goods expensive. This reconciles low real capital accumulation with apparently similar nominal investment rates.

7.3 Sector-level productivity

- Services TFP declines by less than 30 percent under severe financial frictions.
- Manufacturing TFP declines by more than 50 percent.
- The gap arises because manufacturing establishments are larger and harder to self-finance.

Economic message: The sector with larger financing needs is disproportionately hurt by financial frictions.

7.4 Decomposition of sectoral TFP losses

- In services, most TFP loss is due to capital misallocation among active establishments.
- In manufacturing, more than half of the TFP loss comes from misallocation of entrepreneurial talent.
- Talent misallocation includes delayed entry by talented poor individuals and continued operation by low-talent wealthy entrepreneurs.

Interpretation: Capital misallocation is not the whole story. Entry and selection distortions are critical in large-scale sectors.

7.5 Establishment-size predictions

- Financial frictions generate too few entrepreneurs and too large establishments in manufacturing.
- They generate too many entrepreneurs and too small establishments in services.
- The Mexico–United States comparison supports this prediction.

Economic message: The model has a sharp micro prediction: financial underdevelopment changes the relative establishment-size distribution across sectors.

8 Tables and figures

8.1 Figure 1 and Table 1: motivating facts

Read:

- Figure 1 shows that the relative price of manufacturing to services falls with output per worker.
- The right panel shows that relative manufacturing-service TFP rises with output per worker in the OECD sample.
- Table 1 shows that manufacturing is much larger-scale than services: 47 versus 14 workers per establishment in U.S. data.
- Manufacturing also has higher external dependence and a larger capital share.

Economic message: Manufacturing is both larger-scale and more financially dependent. This is the empirical foundation for the model’s two-sector structure.

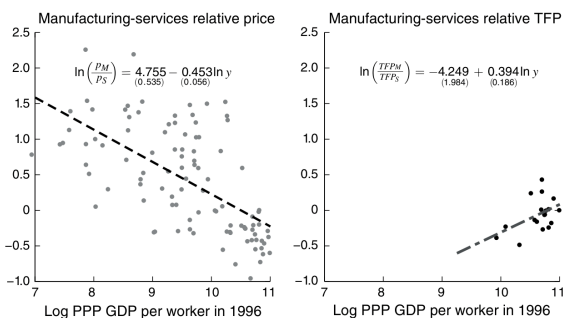


FIGURE 1. RELATIVE PRICES AND RELATIVE PRODUCTIVITY AGAINST OUTPUT PER WORKER

TABLE 1—MANUFACTURING VERSUS SERVICES				
	Workers per establishment	Workers per enterprise	External dependence	Capital share
	US	OECD	US	US
Manufacturing	47	28	0.21	0.31
Services	14	8	0.09	0.27

Figure 1: Relative prices and relative productivity

Table 1: Manufacturing versus services

8.2 Table 2 and Figure 2: calibration and U.S. establishment distribution

Read:

- Table 2 lists calibration targets and parameters.
- The model matches employment concentration, earnings concentration, average sectoral establishment sizes, exit rates, manufacturing share, and the real interest rate.
- Figure 2 compares the calibrated perfect-credit model to the U.S. establishment-size distribution.

- The model matches the tails and the distance between manufacturing and service establishment-size distributions.

Economic message: The calibration disciplines the model with U.S. micro data before using the model to analyze financial development.

TABLE 2—CALIBRATION

Target Moments	US Data	Model	Parameter
Top 10-percentile employment share	0.69	0.69	$\eta = 4.84$
Top 5-percentile earnings share	0.30	0.30	$\alpha + \theta = 0.79$
Average scale in services	14	14	$\kappa_S = 0.00$
Average scale in manufacturing	47	47	$\kappa_M = 4.68$
Establishment exit rate	0.10	0.10	$\gamma = 0.89$
Manufacturing share of GDP	0.25	0.25	$\psi = 0.91$
Interest rate	0.04	0.04	$\beta = 0.92$

Table 2: Calibration



FIGURE 2. ESTABLISHMENT SIZE DISTRIBUTION IN THE MODEL AND DATA

Figure 2: Establishment size distribution

8.3 Figure 3: aggregate effects of financial frictions

Read:

- The left panel plots output per worker relative to the U.S. against external finance to GDP.
- The middle panel plots aggregate TFP relative to the U.S.
- The right panel plots the capital-to-output ratio.
- Model simulations reproduce a substantial part of the observed relationship between finance, output, and TFP.

Economic message: Financial frictions lower aggregate productivity and output. The productivity effect is quantitatively large enough to explain a substantial share of development gaps.

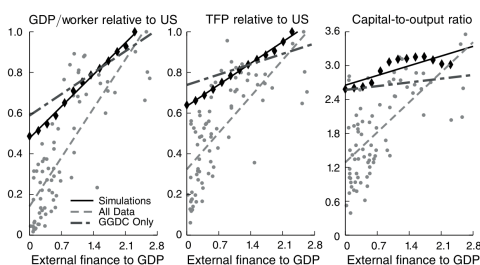


FIGURE 3. AGGREGATE IMPACT OF FINANCIAL FRICTIONS

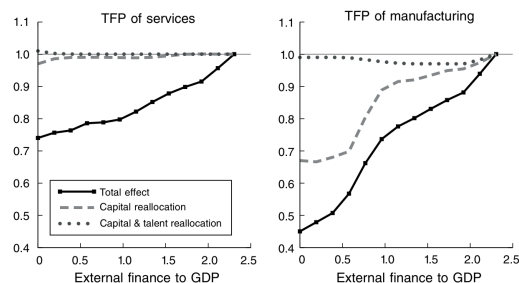


FIGURE 4. IMPACT OF FINANCIAL FRICTIONS ON SECTOR-LEVEL TFP

8.4 Figure 4: sector-level TFP decomposition

Read:

- The service-sector TFP loss is relatively small and mostly reflects capital misallocation.
- The manufacturing-sector TFP loss is much larger.
- In manufacturing, talent misallocation and distorted entry/exit account for more than half of the TFP loss.

- Reallocation of both capital and talent largely restores sectoral TFP, especially in manufacturing.

Economic message: Large-scale sectors are damaged most by financial frictions because productive entrepreneurs cannot easily self-finance the fixed and variable costs needed to operate efficiently.

8.5 Figure 5: relative prices and relative productivity versus finance

Read:

- The left panel plots relative prices of manufacturing to services against external finance to GDP.
- The right panel plots relative manufacturing-service TFP against external finance to GDP.
- The model generates a negative relationship between relative manufacturing prices and financial development.
- The model generates a positive relationship between relative manufacturing productivity and financial development.

Economic message: Sectoral price and productivity patterns are natural consequences of financially constrained large-scale production.

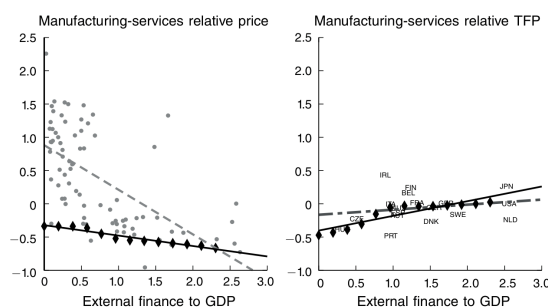


FIGURE 5. RELATIVE PRICES AND RELATIVE PRODUCTIVITY AGAINST EXTERNAL FINANCE

8.6 Figure 6 and Figure 7: establishment-level evidence

Read:

- Figure 6 shows how financial frictions affect average entrepreneurial talent, talent dispersion, average establishment size, and establishment-size dispersion by sector.
- Manufacturing becomes more distorted because high-talent entrepreneurs are constrained and entry is delayed.
- Figure 7 compares industry establishment sizes in Mexico and the United States.
- The relationship supports the prediction that large-scale manufacturing industries have relatively larger establishments in Mexico, while many service industries have smaller establishments.

Economic message: The establishment-level evidence provides a micro validation of the model: financial frictions affect not only aggregate productivity but also the distribution of firms across industries and sectors.

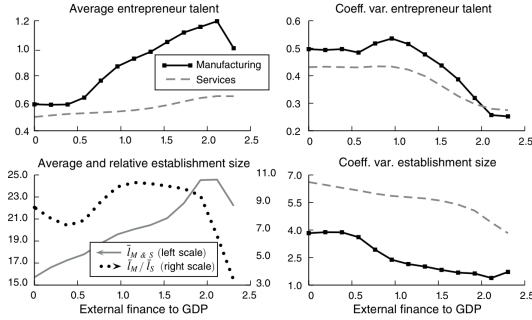


FIGURE 6. ESTABLISHMENT-LEVEL IMPACT OF FINANCIAL FRICTIONS

Figure 6: Establishment-level impact

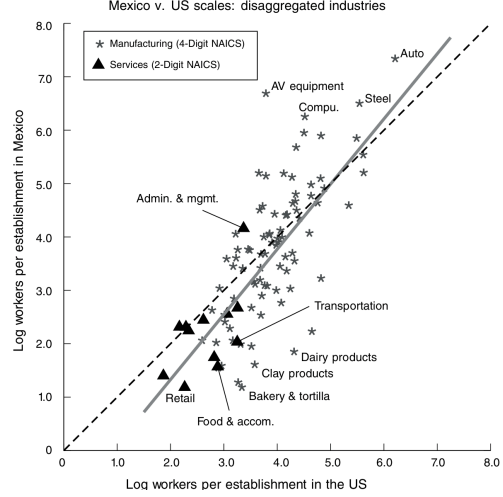


Figure 7: Mexico versus U.S. establishment sizes

9 Short summary

- The paper studies how financial frictions affect development in a two-sector economy.
- Manufacturing is modeled as a large-scale sector with high fixed costs; services are modeled as small scale.
- Financial frictions are modeled as endogenous collateral constraints caused by limited contract enforcement.
- Entrepreneurs differ in wealth and talent, and talent evolves stochastically.
- Wealth accumulation allows self-financing, but self-financing is harder in large-scale sectors.
- The model is calibrated to U.S. establishment-size distributions and macro moments.
- Lower financial development reduces output, aggregate TFP, and capital-output ratios.
- Manufacturing TFP falls more than service TFP because manufacturing has larger financing needs.
- Financial frictions generate high relative prices of manufacturing goods to services in poor countries.
- The model decomposes productivity losses into capital misallocation, talent misallocation, and distorted entry.
- Mexico–United States establishment-size evidence supports the model’s prediction about sectoral scale distortions.

10 Conclusion

10.1 Core conclusion

Buera, Kaboski, and Shin show that financial frictions can account for a substantial part of cross-country differences in output, TFP, sectoral productivity, relative prices, and capital-output ratios. The key is that financial frictions distort both capital allocation and entrepreneurial selection.

10.2 Economic intuition

Large-scale production requires more financing. When financial markets are weak, talented but poor individuals cannot enter or operate manufacturing establishments at efficient scale. They either wait and self-finance, work for wages, or operate too small. Wealthy but low-talent individuals can survive because

they have internal funds. The result is severe misallocation in manufacturing, higher prices of manufactured goods, and lower aggregate productivity.

10.3 Incremental contribution takeaway

The paper's incremental contribution is to connect financial development, sectoral scale, and entrepreneurial selection in one quantitative framework. It goes beyond one-sector models by showing why finance matters differently across sectors and beyond static credit-constraint models by emphasizing self-financing and dynamic wealth accumulation.

10.4 Takeaway for readers

The paper teaches that financial development is not just about raising investment. It changes who becomes an entrepreneur, which sector they enter, how large establishments become, and how productive the economy is. Sectoral scale differences are therefore central to understanding finance and development.